

Alcohol Attenuates CRF-Induced Excitatory Effects from the Extended Amygdala to Dorsostriatal Cholinergic Interneurons

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This **important** study identifies a novel CRF-positive projection from the central amygdala and BNST to dorsal striatal cholinergic interneurons, revealing a previously unrecognized pathway by which stress signals modulate striatal function. The authors present strong and **convincing** evidence for the anatomical and functional connectivity of this circuit and demonstrate that alcohol disrupts CRF-mediated cholinergic activity, supporting its relevance to alcohol use disorder.

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Abstract

Alcohol relapse is linked to corticotropin-releasing factor (CRF) signaling and is caused by dysfunction within reward pathway circuitry, yet the underlying mechanisms guiding this process remain unclear. Here, we investigated how CRF modulates cholinergic interneurons (CINs) in the dorsal striatum, a region critical for mediating cognitive flexibility and action selection. Using monosynaptic and retrograde circuit tracing, we identified direct inputs from CRF-expressing (CRF⁺) neurons in the central amygdala (CeA) and bed nucleus of the stria terminalis (BNST) to dorsal striatal CINs. We showed that CINs express CRF receptor 1 (CRFR1) and established their functional connectivity with CeA/BNST CRF⁺ projections. Functional recordings revealed that CRF enhanced CIN excitability and promoted acetylcholine release in the dorsal striatum. However, acute alcohol exposure and withdrawal attenuated the excitatory effect of CRF on CIN firing, suggesting a mechanism by which alcohol disrupts CRF-dependent neuromodulation. These findings uncover a novel CRF-mediated circuit linking the extended amygdala to the dorsal striatum and provide insight into how CRF and alcohol interact to impair striatal function. This work highlights CRF

signaling as a potential target for uncovering the mechanism of stress-induced changes to the reward pathway.

Highlights

- Dorsal striatal CINs receive monosynaptic CRF⁺ inputs from CeA and BNST neurons.
- CRFR1 is expressed in striatal CINs, and CRF⁺ fibers are present in the dorsal striatum.
- CRF enhances dorsal striatal CIN activity via CRFR1 signaling.
- Acute alcohol exposure impairs CRF-induced cholinergic activity.

Significance Statement

The dorsal striatum regulates goal-directed behavior and is implicated in alcohol use disorder (AUD). Within this region, cholinergic interneurons (CINs) support cognitive flexibility and receive input from limbic areas, including the central amygdala (CeA) and bed nucleus of the stria terminalis (BNST). In this study, we identified direct projections from CRF-producing neurons in the CeA and BNST to dorsal striatal CINs, a subset of which express CRF receptor 1 (CRFR1). Electrophysiological recordings confirmed these projections provide functional input that is disrupted by acute alcohol exposure. These findings lay the groundwork for future studies on how CRF and alcohol interact to impair striatal function.

Introduction

Alcohol use disorder (AUD) is a chronic, recurring brain condition characterized by compulsive alcohol consumption, impaired control over drinking, and negative emotional states during withdrawal. Stress is a key contributor to both the development and recurrence of AUD, with stressful experiences and elevated stress hormone levels frequently precipitating relapse episodes in abstinent individuals [1–3]. An expanding body of clinical and preclinical evidence highlights the importance of stress-responsive neurocircuits in driving addiction-related behaviors, implicating neuropeptides like corticotropin releasing factor (CRF) in relapse vulnerability [4–8]. CRF is a central regulator of the stress response, coordinating hormonal and behavioral adaptations to stress through widespread action in both hypothalamic and extrahypothalamic brain regions. In addition to initiating hypothalamic-pituitary-adrenal (HPA) axis activity, CRF modulates affective and motivational processes via its action in the amygdala, bed nucleus of the stria terminalis (BNST), and other limbic structures.

Substantial evidence indicates that CRF signaling enhances drug-seeking behavior during stress and withdrawal across multiple substances of abuse, supporting its role as a critical mediator of relapse [5, 6, 9]. Despite extensive research on CRF in limbic areas, its role in the dorsal striatum—an area crucial for goal-directed learning, habit formation, and behavioral flexibility—remains less well understood [10–16]. Within this region, cholinergic interneurons (CINs) are key regulators of striatal output and acetylcholine (ACh)-mediated dopamine modulation, integrating diverse inputs and contributing to reward-based learning [17–19]. CINs are sensitive to neuromodulatory influences, yet it is unclear whether they are directly targeted by CRF and how this interaction might be altered by alcohol exposure [20–22].

In this study, we investigated a novel CRF–CIN circuit linking the CeA and BNST to the dorsal striatum. Using monosynaptic and retrograde circuit tracing, we identified direct projections from CRF-expressing neurons to dorsal striatal CINs. CRF enhanced CIN excitability and promoted ACh release via CRFR1 receptors, but this excitatory effect was disrupted by acute alcohol exposure, indicating that alcohol interferes with CRF-dependent cholinergic modulation. These findings identify a CRF-CIN circuit that is vulnerable to alcohol-induced dysregulation, providing mechanistic insight into how stress peptides and alcohol interact to impair striatal function.

Materials and methods

Animals

Male and female 3–4-month-old mice or rats were used in all studies. ChAT-eGFP (stock 007902), ChAT-Cre (stock 031661), *Drd1a*-tdTomato (D1-tdT, stock 016204), Ai32 (stock 012569), CRH-ires-CRE (stock 012704) and C57BL/6J (stock 000664) mice were purchased from The Jackson Laboratory [23–27]. All mice were backcrossed onto a C57BL/6 background. ChAT-Cre mice were crossed with D1-tdTomato mice to obtain ChAT-Cre;D1-tdTomato mice. CRF-Cre mice were crossed with ChAT-eGFP mice to generate CRF-Cre;ChAT-eGFP mice. CRF-Cre;Ai32 mice were generated in-house and crossed with ChAT-eGFP to generate triple transgenic CRF-Cre;Ai32;ChAT-eGFP mice. We used CRFR1-Cre-2A-tdTomato rats that have been validated for expression of Cre and tdTomato in CRFR1-expressing neurons [28]. We also used CRF-Cre rats that have been shown to express Cre recombinase in CRF-producing neurons in the CeA and BNST [29]. We crossed CRF-Cre rats with a Cre-dependent tdTomato reporter line to visualize CRF⁺ neurons. Genotypes were confirmed through PCR analysis of tail DNA to detect Cre or fluorescent protein genes in mice and rats (Cre for CRF-Cre, and ChAT-Cre; tdTomato for D1-tdTomato; and GFP for Ai32) [30–34]. Animals were housed in a temperature- and humidity-controlled vivarium with a 12-h light/dark cycle. Food and water were available *ad libitum*. The Texas A&M University Institutional Animal Care and Use Committee approved all animal care and experimental procedures.

Reagents

AAV8-Ef1a-FLEX-TVA-mCherry (lot # AV5008b), AAV8-FLEX-RG (lot # AV5005f), and AAV-FLEX-Chrimson-tdTomato (lot # AV5844) were purchased from the UNC Vector Core [35, 36]. AAVrg-pCAG-FLEX-eGFP-WPRE (catalog # 51502-AAVrg) was purchased from Addgene [37], while EnvA-SADAG-GFP was purchased from the Salk Institute [38]. Choline acetyltransferase (AB144P) antibody was purchased from Sigma [39]. ACh sensor (AAV-GRABACH4m) was obtained from BrainVTA [40]. CRF peptide and NBI 35695 (CRFR1 antagonist) were obtained from Tocris. 6,7-dinitroquinoxaline-2,3-dione (DNQX), tetrodotoxin (TTX), 4-aminopyridine (4-AP), and bicuculline were also purchased from Tocris.

Stereotaxic virus infusion

The stereotaxic virus infusion procedure was conducted as described previously [30, 41, 42]. When required for the experimental design, AAV-DIO-TVA-mCherry and AAV-DIO-RG were bilaterally infused into the dorsal striatum (AP: 0.38, mm, ML: ±1.55 mm, DV: -2.90 mm from the Bregma) [43] of ChAT-Cre;D1-tdTomato mice (Fig. 1). Rabies-GFP was infused at the same injection site three weeks later at a 10° angle to avoid contamination of the infusion tract. Rabies-GFP virus was allowed to incubate for one week. AAVrg-pCAG-FLEX-eGFP-WPRE was infused into the dorsal striatum (AP: 0.00, mm, ML: ±2.80 mm, DV: -4.85 mm from the Bregma) [44, 45] of CRF-Cre;tdTomato rats (Fig. 3A–3E). Additionally, AAV-DIO-Chrimson-tdTomato was infused into the CeA (AP: -2.60 mm; ML: ±4.20 mm; DV: -8.10 mm) and BNST (AP: -0.10 mm; ML: ±1.40 mm; DV: -6.70 mm) as described in published literature [46] (Fig. 3F–3H). The animals were placed on a stereotaxic surgical frame after being sedated with 3–4% isoflurane at a rate of 1.0 L/min, as described previously [41, 42, 47]. These coordinates were obtained from

previous publications and verified using the rat [48] or mouse [49] brain atlases. A volume of 0.5 $\mu\text{L}/\text{site}$ (mice) or 1 $\mu\text{L}/\text{site}$ (rats) of virus was infused at a rate of 0.08 $\mu\text{L}/\text{min}$. At the end of the infusion, the injectors remained at the injection site for 10 to 15 min before removal to allow for virus diffusion. The scalp incision was then sutured and animals were returned to their home cage for recovery.

Histology and cell counting

Animals were anesthetized and perfused intracardially with 4% paraformaldehyde (PFA) in phosphate-buffered saline (PBS). The brains were then extracted and submerged in 4% PFA/PBS solution for one day at 4°C, then transferred to a 30% sucrose solution in PBS. Once the brains completely sank in the sucrose solution, they were cut into 50- μm thick coronal sections using a cryostat. The slices were stored in a PBS bath at 4°C before mounting on slides for imaging using a confocal laser-scanning microscope (Fluoview, Olympus). All images were processed using Imaris 8.3.1 (Bitplane, Zurich, Switzerland) as previously reported [30, 32, 34]. Immunostaining for choline acetyltransferase (ChAT) was performed using an anti-ChAT primary antibody (AB144P), followed by a fluorophore-conjugated secondary antibody (A21447) emitting at 647 nm, labeling CINs with far-red fluorescence. Cell counting was performed in 8 CRF-Cre;tdTomato rats. For each brain region, 5-10 brain sections were imaged from each animal. Imaris was used to quantify green and red neurons as well as evaluate colocalization. Brain regions were identified using the Mouse Brain Atlas [49].

Slice electrophysiology

Slice preparation

Slices were prepared and electrophysiological recordings were conducted as described previously [30, 32, 41, 42, 50]. Briefly, coronal sections (250 μm) containing the striatum were cut in an ice-cold cutting solution containing (in mM): 40 NaCl, 148.5 sucrose, 4 KCl, 1.25 NaH_2PO_4 , 25 NaHCO_3 , 0.5 CaCl_2 , 7 MgCl_2 , 10 glucose, 1 sodium ascorbate, 3 sodium pyruvate, and 3 myo-inositol. The solution was saturated with 95% O_2 and 5% CO_2 . Slices were then incubated in a 1:1 mixture of the cutting and external solutions at 32°C for 45 min. The external solution was composed of the following (in mM): 125 NaCl, 4.5 KCl, 2.5 CaCl_2 , 1.3 MgCl_2 , 1.25 NaH_2PO_4 , 25 NaHCO_3 , 15 glucose, and 15 sucrose. The external solution was saturated with 95% O_2 and 5% CO_2 . Slices were then maintained in the external solution at room temperature until use.

Cell-attached recordings

Individual slices were transferred to a recording chamber and continuously perfused with the external solution at 2-3 mL/min at 32°C. Fluorescent axonal fibers and neurons were visualized using an epifluorescent microscope (Olympus). The presence of tdTomato and GFP allowed cell type verification and expression of Ai32 allowed visualization of CRF-producing fibers. CINs in slices were identified by their labeled color, large size, and spontaneous firing. Spontaneous cell-attached CIN firing activity was recorded for 5 min to calculate the average firing frequency. 470-nm or 590-nm light was emitted from the objective lens for 2 ms at a frequency and duration specified in the respective figure legend to stimulate axonal fibers containing channelrhodopsin in the dorsal striatum. We used a potassium-based intracellular solution containing (in mM): 123 potassium gluconate, 10 HEPES, 0.2 EGTA, 8 NaCl, 2 MgATP, and 0.3 NaGTP, with an osmolarity of ~ 280 mOsm/L and the pH adjusted to 7.3 using KOH. For protocols involving alcohol application, we pretreated dorsal striatal slices with alcohol (50 mM) for 1 h outside of the recording chamber. After pretreatment, slices were transferred to the recording chamber, where alcohol was washed out for at least 15 min before recording. The recordings included a 10-min baseline period (without alcohol), followed by 10 min of CRF bath application, and then 10 min of alcohol exposure. Control slices did not receive alcohol pretreatment but were exposed to acute alcohol

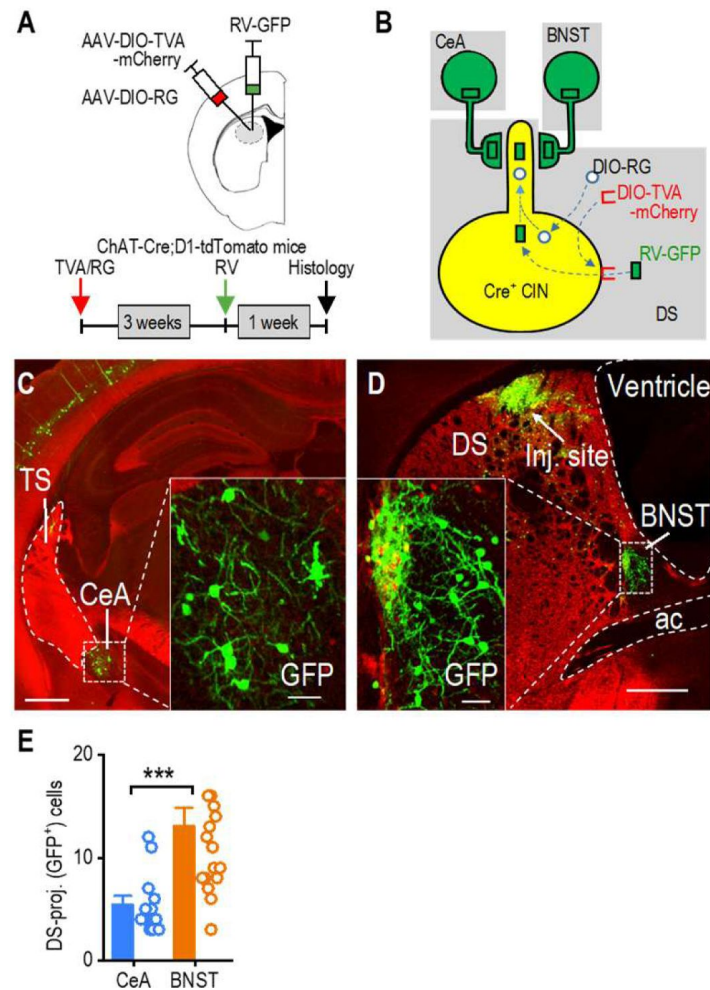


Figure 1.

Dorsal Striatal CINs Receive Monosynaptic Inputs From the CeA and BNST

A, Schematic illustrating the infusion of helper viruses (AAV-DIO-TVA-mCherry and AAV-DIO-RG) and rabies-GFP (RV-GFP) into the dorsal striatum of ChAT-Cre;D1-tdTomato mice. The rabies-GFP was infused 3 weeks after the helper virus infusion, and animals were euthanized one week after the rabies infusion. **B**, Model for the retrograde transsynaptic labeling of CeA and BNST neurons projecting to dorsal striatal CINs. TVA permits selective infection by the rabies-GFP virus, while RG mediates the retrograde transsynaptic jump from postsynaptic CINs to presynaptic terminals of CeA or BNST neurons. **C**, Sample image showing CeA neurons projecting to dorsal striatal CINs, as indicated by rabies-GFP expression. AP: -1.50 mm from Bregma. TS, tail of the striatum. Scale bar: 0.5 mm, 50 μ m for insert. **D**, Sample image demonstrating that BNST neurons project to dorsal striatal CINs, as indicated by rabies-GFP expression. Note that the injection site (inj. site) is also displayed in the dorsal striatum. AP: -0.26 mm from Bregma. DS, dorsal striatum; ac, anterior commissure. Scale bar: 0.5 mm, 50 μ m for insert. **E**, The CeA and BNST both send projections to the dorsal striatum, with the BNST providing more inputs. *** $p < 0.001$ by Mann-Whitney test. $n = 13$ sections from 4 mice (13/4) for the CeA and 18/4 for the BNST.

following CRF application. In both conditions, the final 10-min alcohol application allowed us to assess whether alcohol pretreatment altered the CIN firing response to subsequent CRF and alcohol exposure. In protocols that required inhibition of glutamatergic and GABA_A receptors, DNQX and bicuculline were used, respectively. Electrophysiology data were acquired using Clampex-10 (Molecular Devices) and analyzed using Clampfit-10 (Molecular Devices) and Mini Analysis (Mini60, Synaptosoft Inc.).

Ex vivo live-tissue confocal imaging of ACh release

Brain slices were kept in a recording chamber perfused with external solution flowing and saturated with 95% O₂ and 5% CO₂. An Olympus FluoView FV3000 microscope was used with a 10x NA 0.3 and a 40x NA 0.8 water immersion objective, along with a 488 nm and a 561 nm laser. The sample rate of imaging was 2-3 frames per second. Parameters were maintained consistently across all imaging sessions, including laser intensity, HV, gain, offset, and aperture diameter.

Statistical analysis

Before conducting all experiments shown in this study, we performed a power analysis with SigmaPlot software (12.5, Systat) using the mean and standard deviation from previous studies in our lab [34, 43–45] to determine the required sample sizes to detect a significant difference. We tested all data for normality before significance testing. If the normality test failed, we used nonparametric tests, such as the Mann-Whitney U Test. All data are expressed as the mean ± the standard error of the mean. Data were analyzed by two-tailed *t*-test (unpaired or paired), one- or two-way ANOVA with repeated measurement, followed by the Student-Newman-Keuls (SNK) post hoc test. Significance was determined if *p* < 0.05. Statistical analysis was conducted by the SigmaPlot program. Graphs were constructed using the OriginPro (2024b, OriginLab) program.

Results

Dorsal Striatal CINs Receive Monosynaptic Inputs From CeA and BNST Neurons

To investigate the connection between dorsal striatal CINs and stress-related brain regions, we examined whether CINs receive monosynaptic inputs from the CeA and the BNST [51, 52]. We used ChAT-Cre;D1-tdTomato mice to perform rabies-mediated monosynaptic retrograde tracing. In this mouse model, CINs express Cre recombinase and dopamine D1-receptor (D1R)-expressing medium spiny neurons (D1-MSNs) are labeled with tdTomato. Including the D1-tdTomato marker allowed us to delineate the CeA and BNST, which do not express D1Rs unlike surrounding striatal areas [47]. We performed rabies-mediated retrograde monosynaptic circuit tracing (Fig. 1A), allowing specific targeting of CINs (Fig. 1B). In addition to labeling neurons in brain regions known to project to dorsal striatal CINs, we were surprised to see labeled neurons in both the CeA (Fig. 1C) and BNST (Fig. 1D), which are well-characterized as key stress-responsive nuclei. Interestingly, we observed significantly more GFP⁺ neurons in the BNST than in the CeA (Fig. 1E; Mann-Whitney U = 29.500, ****p* < 0.001). These results demonstrate that dorsal striatal CINs receive direct, monosynaptic inputs from neurons in the CeA and BNST. Together, these findings highlight the anatomical connection of dorsal striatal CINs with key stress-responsive brain regions.

The Dorsal Striatum Lacks CRF⁺ Neurons Unlike the CeA and BNST

We first investigated whether the dorsal striatum contains CRF producing cells or if CRF signaling in the dorsal striatum instead depends on CeA and BNST inputs. While previous studies have identified CRF-producing neurons in the CeA and BNST [46], their presence in the dorsal

striatum remains uncertain. Coronal brain sections from CRF-Cre;tdTomato rats revealed dense populations of CRF-tdTomato⁺ neurons in the CeA (**Fig. 2A**) and BNST (**Fig. 2B**). Although CRF⁺ axonal fibers were observed in the dorsal striatum, CRF⁺ cell bodies were largely absent (**Fig. 2C**). Quantitative analysis confirmed significant regional differences, indicating that the CeA and BNST contained substantially more CRF⁺ neurons than the dorsal striatum (**Fig. 2D**; CeA versus DS, $Q = 7.30$, $*p < 0.05$; BNST versus DS, $Q = 3.72$, $*p < 0.05$). Notably, the CeA exhibited a higher number of CRF⁺ neurons compared with the BNST (**Fig. 2D**; $Q = 2.75$, $*p < 0.05$). The apparent lack of CRF⁺ cell bodies in the dorsal striatum suggests that CRF signaling in the dorsal striatum arises from CRF⁺ neurons in the CeA and BNST.

Dorsal Striatum CINs Receive Monosynaptic

CRF⁺ Inputs from the CeA and BNST

To test whether CRF⁺ neurons in the CeA and BNST project to the dorsal striatum, we infused a retrograde adeno-associated virus into the dorsal striatum of CRF-Cre;tdTomato rats. This strategy allows CRF⁺ neurons (tdTomato⁺) that project to the dorsal striatum to also be labeled with GFP (**Fig. 3A**). Confocal imaging revealed GFP⁺ neurons in both the BNST (**Fig. 3B**) and CeA (**Fig. 3C**), with many overlapping GFP-labeled dorsal striatum-projecting neurons and tdTomato-positive (CRF⁺) cells in both regions. This indicates that CRF⁺ neurons from the CeA and BNST project to the dorsal striatum. Despite the CeA containing more CRF⁺ neurons than the BNST, as we observed above, the proportion of dorsal striatum-projecting CRF⁺ neurons was higher in the BNST than in the CeA (**Fig. 3D**, **3E**; $t_{15} = 5.49$, $***p < 0.001$; $t_{15} = -5.01$, $***p < 0.001$). However, the total number of dorsal striatum-projecting CRF⁺ neurons did not differ between the CeA and BNST (**Fig. 3F**; $t_{15} = -0.78$, $p > 0.05$). These results indicate that CRF⁺ neurons in the CeA and BNST project to the dorsal striatum.

To test whether CRF neurons in the CeA and BNST make monosynaptic projections to dorsal striatal CINs, we performed ChR2-assisted circuit tracing. Given that both the CeA and BNST are primarily GABAergic nuclei [53, 54], we examined whether optogenetic stimulation of CRF-containing projections to the dorsal striatum induce inhibitory monosynaptic currents in CINs. To test this, we infused AAV-FLEX-Chrimson-tdTomato into the CeA and BNST of CRF-Cre;ChAT-eGFP mice so that we could selectively stimulate CRF-containing fibers in the dorsal striatum while recording from GFP labeled CINs (**Fig. 3G**). We found that optogenetic stimulation of CRF⁺ fibers with yellow light evoked fast synaptic currents in dorsal striatal CINs (**Fig. 3H**). This response was abolished by TTX (**Fig. 3H**, **3I**; BL versus TTX: $q = 4.16$, $*p < 0.05$) but restored using TTX+4-AP (**Fig. 3H**, **3I**; TTX vs TTX+4-AP: $q = 4.73$, $*p < 0.05$), suggesting the existence of monosynaptic transmission. Notably, the restored synaptic response was predominantly blocked by the GABA_A receptor antagonist bicuculline (**Fig. 3H**, **3I**; TTX+4-AP versus TTX+4-AP+Bic: $q = 4.44$, $*p < 0.05$), demonstrating that evoked currents are GABAergic. Together, these results demonstrate that dorsal striatal CINs receive monosynaptic GABAergic synapses from the CRF⁺ neurons in the CeA and BNST, suggesting that CRF⁺ neurons project to the dorsal striatum and release CRF to modulate cholinergic signaling in the dorsal striatum.

Striatal CINs Express CRFR1

Having established that CINs receive monosynaptic inputs from CRF neurons in the CeA and BNST, we next replicated the finding that CINs express the CRF receptor, CRFR1 [15, 55, 56]. To address this, we used CRFR1-Cre-tdTomato rats [28], in which CRFR1-containing neurons express Cre recombinase and are visible due to the tdTomato label. Confocal imaging of striatal sections from these rats confirmed that CRFR1-expressing neurons are distributed throughout the striatum (**Fig. 4A**). To verify that CRFR1-positive (CRFR1⁺) neurons are also cholinergic, we stained sections with an anti-ChAT antibody and imaged them using confocal microscopy (**Fig.**

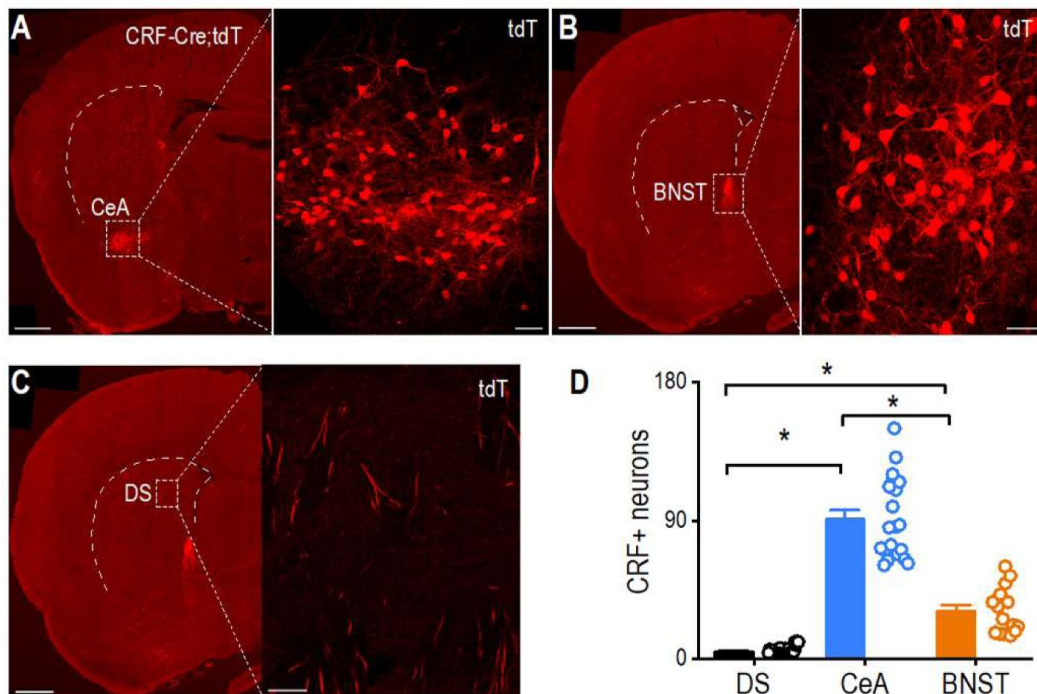


Figure 2.

CRF-Positive Neurons Are Abundant in the CeA and BNST but Absent in the Dorsal Striatum

A, B, Confocal images showing dense populations of tdTomato-labeled CRF⁺ neurons in the CeA (A) and BNST (B) of CRF-Cre;tdTomato rats. AP: -1.5 mm from Bregma (A) and -0.26 mm (B). Scale bars: 1 mm (left) and 50 μ m (right). DS, dorsal striatum. **C,** Representative image of the dorsal striatum showing the presence of CRF⁺ axonal fibers but the absence of CRF⁺ cell bodies. AP: -0.26 mm from Bregma. Scale bar: 100 μ m. **D,** The CeA and BNST contain more CRF⁺ neurons than the dorsal striatum. * $p < 0.05$ by Kruskal-Wallis with Dunn's Method. $n = 20$ slices from 7 rats (20/7) for the CeA, 29/7 for the striatum, and 15/7 for the BNST.

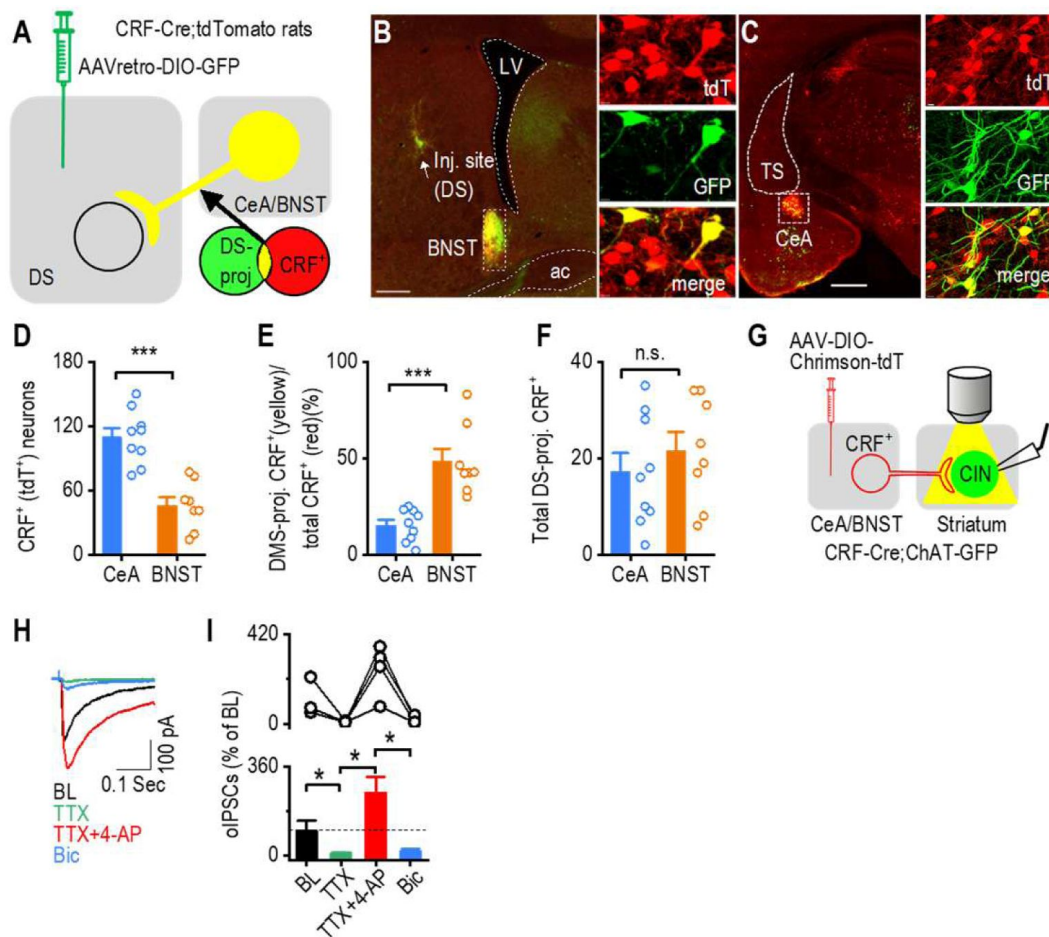


Figure 3.

CINEs Receive Monosynaptic Input From CeA and BNST CRF⁺ Neurons

A, Schematic showing virus injection (AAVretro-DIO-GFP) into the dorsal striatum of CRF-Cre;tdTomato rats. **B**, Representative image showing the injection site of AAVretro-DIO-GFP in the posterior dorsal striatum of CRF-Cre;tdTomato rats. Overlapping tdTomato⁺ (CRF⁺) and dorsal striatum-derived Cre-driven GFP⁺ expression in the BNST (depicted in yellow) indicates that these CRF⁺ neurons project to the dorsal striatum. ac, anterior commissure; LV, left ventricle; tdT, tdTomato; DS-proj, dorsal-striatum-projecting. Scale bar: 0.5 mm (left), 10 μ m (right). **C**, Images showing the overlap (in yellow) of tdTomato⁺ (CRF⁺) and dorsal striatum-derived Cre-driven GFP⁺ expression in the CeA, confirming that these CRF⁺ neurons also project to the dorsal striatum. TS, the tail of the striatum. Scale bar: 1 mm (left), 10 μ m (right). **D**, The CeA contains more CRF⁺ neurons than the BNST. *** p < 0.001 by unpaired t test, n = 9 sections from 5 rats (9/5) for the CeA and 8/5 for the BNST. **E**, The proportion of CRF⁺ neurons projecting to the dorsal striatum is higher in the BNST compared to the CeA. *** p < 0.001 by unpaired t test, n = 9/5 (CeA) and 8/5 (BNST). **F**, The total number of dorsal striatum-projecting CRF⁺ neurons does not differ between these two regions. n.s., not significant, p > 0.05. n = 9/5 (CeA) and 8/5 (BNST). **G**, Schematic showing the injection of AAV-FLEX-Chrimson-tdTomato into the CeA and BNST of CRF-Cre;ChAT-eGFP mice and the subsequent recording of green striatal CINEs during blue light stimulation of surrounding Chrimson-containing fibers from CRF⁺ neurons. Fibers were stimulated at a wavelength of 590 nm for 2 ms. **H**, Sample traces showing CIN responses to blue light stimulation of CRF fibers, which were abolished by TTX, recovered with TTX+4-AP and further eliminated by bicuculline (Bic). **I**, Summary of oIPSC data showing the disappearance of oIPSCs with TTX and reappearance with TTX and 4-AP. * p < 0.05 by one-way ANOVA. n = 4 slices.

4A [4A](#)). We found that ~30% of CINs express CRFR1 (**Fig. 4B** [4B](#), **4C** [4C](#); Mann-Whitney $U = 0.000$, $***p < 0.001$). These findings align with previous reports that a subset of striatal CINs express CRFR1, suggesting that the striatum is involved in CRFR1-mediated signaling pathways.

CRF Enhances CIN Activity and Acetylcholine Release in the Dorsal Striatum

Following the finding that CRF⁺ neurons send inputs to the dorsal striatum, we sought to investigate how CRF might influence CIN spontaneous firing activity. Cell-attached electrophysiology recordings of GFP⁺ neurons in the dorsal striatum were performed in ChAT-eGFP mice before and during bath application of CRF (100 nM) (**Fig. 5A** [5A](#)). We found that CRF significantly increased CIN firing frequency above baseline (**Fig. 5B** [5B](#), **5C** [5C](#); $t_{(8)} = -24.08$, $***p < 0.001$). This effect was abolished by pre-treatment with the CRFR1 antagonist NBI 35695 (**Fig. 5C** [5C](#); Mann-Whitney $U = 0.000$, $***p < 0.001$), confirming that CRF potentiates CIN firing via CRFR1 activation in the dorsal striatum.

Because CINs are the primary source of ACh in the dorsal striatum, we next examined whether CRF-induced CIN activation enhances ACh release. To do so, we infused an adeno-associated virus (AAV) introducing a genetically encoded ACh sensor (AAV-GRAB_{ACh4m}) into the dorsal striatum of wildtype mice [\[40\]](#). Live-tissue confocal imaging was performed on brain slices containing the dorsal striatum 14 days post-infusion to monitor ACh levels (**Fig. 5D** [5D](#)). Bath application of CRF produced a robust ACh fluorescence signal (**Fig. 5E** [5E](#), **5F** [5F](#); Mann-Whitney $U = 7.000$, $*p < 0.05$), indicating that CRF enhances the release of ACh. These results suggest that CRF-driven CIN activation increases ACh release in the dorsal striatum.

To further determine whether direct activation of CRF-expressing fibers modulates CIN activity, we used an optogenetic approach. We generated CRF-Cre;Ai32;ChAT-eGFP mice to selectively stimulate CRF⁺ terminals in the dorsal striatum. Cell-attached electrophysiology recordings were conducted in CINs while blue light (470 nm, 50 Hz, 60 s) was used to activate local ChR2-expressing CRF⁺ fibers (**Fig. 5G** [5G](#)). Burst stimulation of CRF⁺ terminals significantly and reversibly increased CIN firing in the presence of DNQX and bicuculline (**Fig. 5H** [5H](#), **5I** [5I](#); Pre versus Stim, $q = 3.69$, $*p < 0.05$; Stim versus Post, $q = 11.82$, $***p < 0.001$). Together, these findings demonstrate that CRF enhances CIN activity and ACh release in the dorsal striatum.

Acute Alcohol Application Attenuates CRF-Mediated Enhancement of CIN Firing

Given that acute alcohol exposure suppresses CIN activity [\[20\]](#), we next examined the impact of alcohol on CRF-mediated enhancement of CIN firing. We first assessed how alcohol alone influences CIN firing by measuring spontaneous CIN activity before, during, and after bath application of 50 mM alcohol in dorsal striatal slices from ChAT-eGFP mice. Consistent with previous reports [\[20\]](#), acute alcohol suppressed CIN spontaneous firing in cell-attached recordings, an effect that was reversed after a washout period (**Fig. 6A** [6A](#), **6B** [6B](#); Mann-Whitney $U = 0.000$, $*p < 0.05$).

To determine how acute alcohol exposure affects CRF-induced enhancement of CIN firing, we pretreated dorsal striatal slices with alcohol (50 mM) for 1 h outside of the recording chamber (**Fig. 6C** [6C](#)). In control slices, CRF significantly increased the spontaneous firing of dorsal striatal CINs, consistent with our previous findings (**Fig. 6D** [6D](#); BL versus CRF, $q = 11.51$, $***p < 0.001$; see also **Fig. 5B** [5B](#)). Subsequent alcohol application reduced firing but maintained a level still higher than the baseline (**Fig. 6D** [6D](#); CRF versus EtOH, $q = 6.21$, $**p < 0.01$; BL versus EtOH, $q = 5.29$, $**p < 0.01$). This residual increase is likely due to the persistent effect of exogenous CRF, which is sustained for several minutes (**Fig. 5B** [5B](#)).

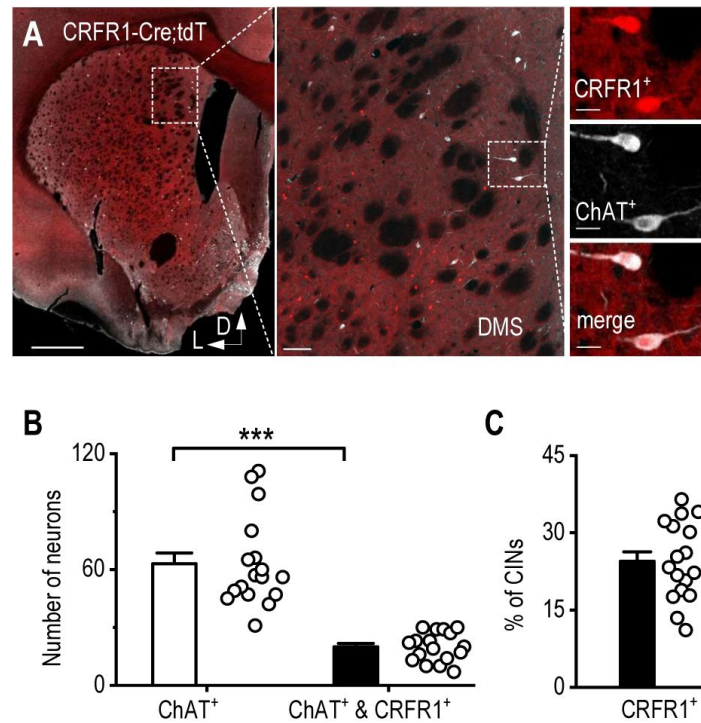


Figure 4.

CRFR1 is Expressed in Striatal CINs

A, Representative image of a coronal striatal section from a CRFR1-Cre-tdTomato rat showing CRFR1 expression overlapping with anti-ChAT immunoreactivity. The section was stained using an anti-ChAT antibody and imaged using a far-red (647 nm) wavelength (white). Scale bars: 1 mm (left), 100 μ m (middle), and 20 μ m (right). **B**, There are significantly fewer CRFR1⁺ CINs than the total number of CINs in the dorsal striatum. *** $p < 0.001$ by Mann-Whitney test. $n = 8$ sections from 2 rats. **C**, Around 30% of CINs express CRFR1. $n = 8$ sections from 2 rats.

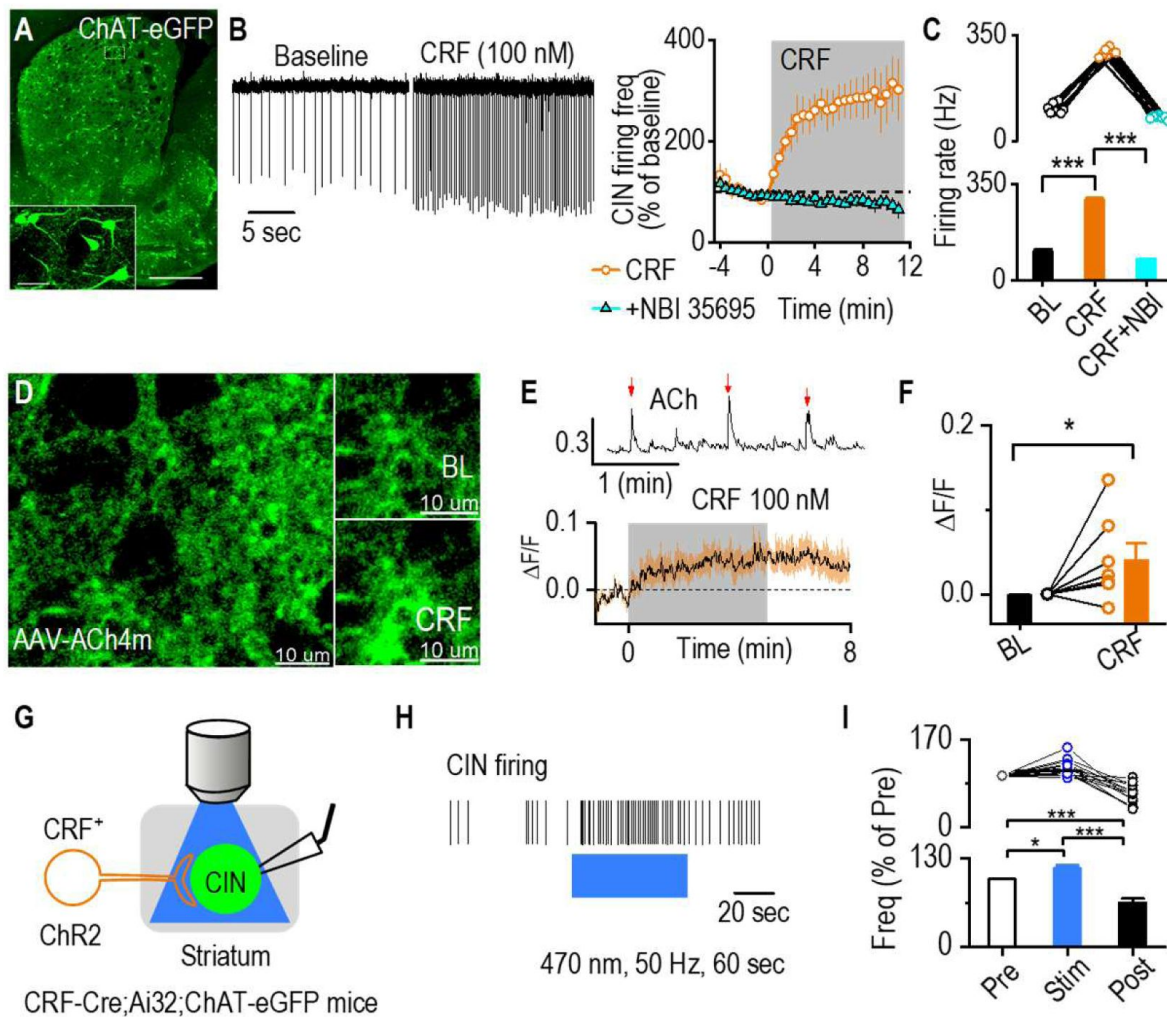


Figure 5.

CRF Enhances CIN Activity and ACh Release in the Dorsal Striatum

A, Sample images of GFP-labeled CINs in the striatum of a ChAT-eGFP mouse. Scale bar: 0.5 mm, 50 μ m (inset). **B**, Bath application of CRF (100 nM) increased the spontaneous firing of dorsal striatal CINs in cell-attached electrophysiological recordings. This effect was prevented by pretreatment with the CRFR1 antagonist NBI 35695 (5 μ M). $n = 7$ cells from 3 mice (7/3) for CRF and 6/3 for CRF plus antagonist recordings. **C**, Data showing that CRF significantly increases the firing from baseline and the firing frequency in the presence of CRF following pretreatment with CRFR1 antagonist is significantly lower. $***p < 0.001$ by paired t-test, $***p < 0.001$ by Mann-Whitney test. $n = 7$ cells from 3 mice. **D**, Sample images of ACh sensor fluorescence in dorsal striatal slices before and during bath application of CRF (100 nM). AAV-GRAB_{ACh4m} was infused into the dorsal striatum of wildtype mice, and live-tissue confocal imaging was conducted two weeks post-infusion. Scale bar: 10 μ m for left and right. **E**, Sample trace of spontaneous ACh release events (indicated by red arrows, top). Bath application of CRF increased ACh sensor fluorescence (bottom). **F**, Summary data showing a significant increase in ACh sensor fluorescence following CRF application. $*p < 0.05$ by Mann-Whitney test. $n = 7$ slices from 7 mice. **G**, Schematic of cell-attached electrophysiological recordings from dorsal striatal CINs in CRF-Cre;Ai32;ChAT-eGFP mice, with simultaneous optogenetic stimulation of CRF⁺ fibers using blue light (470-nm, 2 ms, 50 Hz, 60 sec). **H**, Sample trace showing the increase in CIN firing frequency during blue light stimulation of CRF⁺ fibers. **I**, Data demonstrating a significant and reversible increase in CIN firing frequency during optogenetic stimulation, $*p < 0.05$, $***p < 0.001$ by one-way RM ANOVA. $n = 13$ neurons from 8 animals.

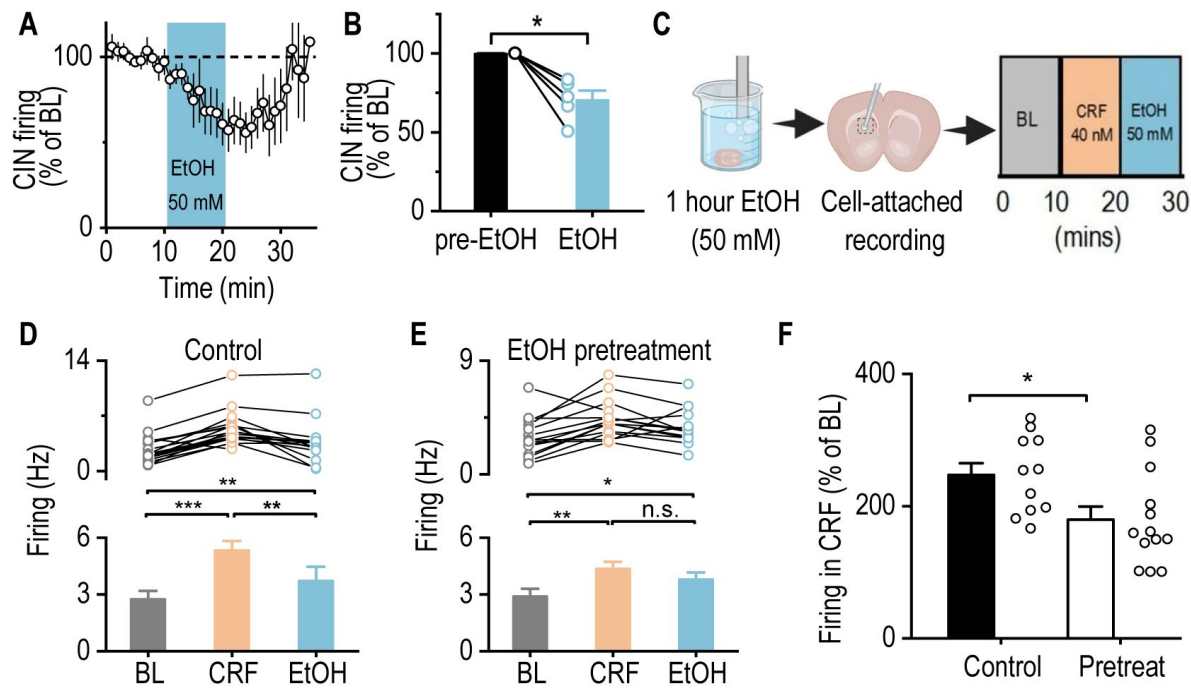


Figure 6.

Acute Alcohol Exposure Attenuates CRF-Induced Enhancement of CIN Activity

A, Time course of spontaneous firing in CINs from dorsal striatal slices of ChAT-eGFP mice, before, during, and after bath application of alcohol (EtOH; 50 mM). **B**, Summary data demonstrates a significant reduction in CIN firing frequency following alcohol application. $**p < 0.01$ by Mann-Whitney test. $n = 6$ cells from 4 mice. **C**, Schematic of the experimental design in which striatal slices were pretreated with alcohol (50 mM) for 1 h in an incubation chamber and then washed for 15 min in the recording chamber. CINs were selected for cell-attached electrophysiological recordings, measuring firing frequency for 10 min (baseline), followed by CRF (40 nM) and alcohol (50 mM) bath applications for 10 min each. **D**, **E**, Data showing spontaneous firing of CINs during baseline (BL), CRF, and alcohol bath application for the control (**D**) and alcohol pretreated (**E**) groups. $*p < 0.05$, $**p < 0.01$, $***p < 0.001$ by one-way RM ANOVA. $n = 10$ cells from 8 mice (10/8) for the control group and 10/7 for the alcohol pretreated group. **F**, Data showing that alcohol pretreatment attenuated CRF-induced enhancement of CIN firing frequency. $*p < 0.05$ by unpaired t-test. $n = 10$ cells from 8 mice (10/8) for the control group and 10/7 for the pretreated group.

In slices pretreated with alcohol, CRF still increased CIN firing frequency, but subsequent alcohol application failed to suppress it back to baseline (**Fig. 6E**; BL versus CRF, $q = 5.20$, $^{**}p < 0.01$; CRF versus EtOH, $q = 6.21$, $p = 0.05$). This suggests that prior acute alcohol exposure reduces the ability of subsequent alcohol exposure to suppress CIN activity in the presence of CRF. Notably, when comparing the change in CIN firing from baseline to CRF application, alcohol pretreatment significantly blunted CRF-induced enhancement of CIN firing (**Fig. 6F**; $t_{18} = 2.70$, $^{*}p < 0.05$). Together, these findings demonstrate that acute alcohol attenuates subsequent CRF-mediated enhancement of CIN firing, suggesting a potential mechanism by which alcohol and CRF interact to modulate striatal cholinergic activity.

Discussion

This study identifies a circuit in which CRF-positive neurons in the CeA and BNST provide direct input to dorsal striatal CINs that express CRFR1. CRF enhances CIN excitability and ACh release, linking CRF to cholinergic modulation. In addition, alcohol exposure and withdrawal blunt this effect, suggesting a possible role for CRF signaling to dorsal striatal CINs in the etiology of alcohol use disorder. Using monosynaptic and retrograde circuit tracing, we identified direct projections from CRF-expressing neurons in the CeA and BNST to dorsal striatal CINs. Functional electrophysiology shows that CRF enhances CIN excitability and promotes ACh release via CRFR1 receptors. However, acute alcohol exposure and withdrawal disrupted this excitatory effect, indicating that alcohol interferes with CRF-dependent cholinergic modulation. These findings identify a CRF–CIN circuit that is vulnerable to alcohol-induced dysregulation, providing mechanistic insight into how CRF and alcohol interact to impair striatal function. Our results highlight CRF signaling in the dorsal striatum as a potential therapeutic target for mitigating the cognitive flexibility deficits and reward system dysfunction seen in AUD. Monosynaptic Inputs from the CeA and BNST to CINs

Tracing experiments reported here show that dorsal striatal CINs receive direct synaptic input from CRF-expressing neurons in the CeA and BNST, regions involved in stress and emotion [53, 57]. While previous studies have identified CeA and BNST projections to the striatum, they did not show that they arise from CRF neurons or that they target CINs [47, 58–60]. Our results establish a direct CRF-to-CIN pathway, linking CRF release to cholinergic modulation. Given CINs' roles in striatal output, behavioral flexibility, and reward learning [45, 61–64], this circuit likely contributes to stress-driven behavioral adaptations [53, 65–68].

We found that there are no CRF neurons in the dorsal striatum, indicating that CRF signaling in the dorsal striatum arises from inputs outside of the striatum, including from the CeA and BNST. Electrophysiological experiments show that CRF fiber stimulation enhances CIN excitability in the presence of synaptic blockers, supporting a role for CRF signaling in the modulation of striatal cholinergic tone. The transient effects of endogenous CRF release, compared to prolonged effects of CRF bath application, suggest that CRF signaling in the striatum may generate dynamic patterns of cholinergic activation to influence behavioral expression.

Alcohol Disrupts CRF-Mediated CIN Excitation

A key finding of this study is that alcohol exposure disrupts CRF-mediated CIN excitation. Both acute alcohol exposure and withdrawal attenuate the CRF-induced increase in CIN activity, indicating that alcohol alters the responsiveness of these neurons to stress signals. Because stress is a well-established trigger for alcohol relapse [69–71], the observed alcohol-induced disruption in CRF signaling may impair the brain's ability to adaptively respond to stress, thereby increasing susceptibility to relapse.

One possibility is that alcohol exposure downregulates CRFR1 expression [72 [↗](#), 73 [↗](#)] or alters CIN function [44 [↗](#)], reducing their responsiveness to stress-related input [74 [↗](#), 75 [↗](#)]. Alternatively, alcohol may interfere with intracellular signaling cascades downstream of CRFR1 activation, leading to diminished excitability and ACh release. The mechanisms underlying alcohol's effect on CRF signaling in CINs remain to be fully elucidated.

Implications for Alcohol Use Disorder

By delineating a CRF-to-CIN pathway linking the extended amygdala to the dorsal striatum, our findings provide critical insights into how stress and alcohol interact at the circuit level to shape behavior. Given the role of CINs in action selection and behavioral flexibility [76 [↗](#)], disruption of this stress-sensitive cholinergic circuit may underlie the decision-making deficits and compulsive alcohol-seeking behavior observed in AUD [43 [↗](#)–45 [↗](#), 77 [↗](#)–79 [↗](#)].

In summary, we identify a direct CRF-positive projection from the CeA and BNST to dorsal striatal CINs, revealing a new mechanism by which stress can modulate cholinergic signaling in the striatum. We demonstrate that CRF enhances CIN excitability and ACh release, and that alcohol exposure disrupts this modulation. These findings highlight the CRF–CIN circuit as a potential therapeutic target in AUD and suggest that restoring CRF signaling may help counteract stress- and alcohol-induced striatal dysfunction. Future work should assess whether pharmacological or circuit-based interventions targeting CRFR1 or CINs can mitigate behavioral impairments in individuals with AUD.

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Reviewer #1 (Public review):

Summary:

The authors show that corticotropin-releasing factor (CRF) neurons in the central amygdala (CeA) and bed nucleus of the stria terminalis (BNST) monosynaptically target cholinergic interneurons (CINs) in the dorsal striatum of rodents. Functionally, activation of CRFR1 receptors increases CIN firing rate, and this modulation was reduced by pre-exposure to ethanol. This is an interesting finding, with potential significance for alcohol use disorders, but some conclusions could use additional support.

Strengths:

Well-conceived circuit mapping experiments identify a novel pathway by which the CeA and BNST can modulate dorsal striatal function by controlling cholinergic tone. Important insight into how CRF, a neuropeptide that is important in mediating aspects of stress, affective/motivational processes, and drug-seeking, modulates dorsal striatal function.

Weaknesses:

- (1) Tracing and expression experiments were performed both in mice and rats (in a mostly non-overlapping way). While these species are similar in many ways, some conclusions are based on assumptions of similarities that the presented data do not directly show. In most cases, this should be addressed in the text (but see point number 2).
- (2) Experiments in rats show that CRFR1 expression is largely confined to a subpopulation of striatal CINs. Is this true in mice, too? Since most electrophysiological experiments are done in various synaptic antagonists and/or TTX, it does not affect the interpretation of those data, but non-CIN expression of CRFR1 could potentially have a large impact on bath CRF-induced acetylcholine release.
- (3) Experiments in rats show that about 30% of CINs express CRFR1 in rats. Did only a similar percentage of CINs in mice respond to bath application of CRF? The effect sizes and error bars in Figure 5 imply that the majority of recorded CINs likely responded. Were exclusion criteria used in these experiments?
- (4) The conclusion that prior acute alcohol exposure reduces the ability of subsequent alcohol exposure to suppress CIN activity in the presence of CRF may be a bit overstated. In Figure 6D (no ethanol pre-exposure), ethanol does not fully suppress CIN firing rate to baseline after CRF exposure. The attenuated effect of CRF on CIN firing rate after ethanol pre-treatment (6E) may just reduce the maximum potential effect that ethanol can have on firing rate after CRF, due to a lowered starting point. It is possible that the lack of significant effect of ethanol after CRF in pre-treated mice is an issue of experimental sensitivity. Related to this point, does pre-treatment with ethanol reduce the later CIN response to acute ethanol application (in the absence of CRF)?

(5) More details about the area of the dorsal striatum being examined would be helpful (i.e., a-p axis).

<https://doi.org/10.7554/eLife.107145.1.sa3>

Reviewer #2 (Public review):

Summary:

Essoh and colleagues present a thorough and elegant study identifying the central amygdala and BNST as key sources of CRF input to the dorsal striatum. Using monosynaptic rabies tracing and electrophysiology, they show direct connections to cholinergic interneurons. The study builds on previous findings that CRF increases CIN firing, extending them by measuring acetylcholine levels in slices and applying optogenetic stimulation of CRF+ fibers. It also uncovers a novel interaction between alcohol and CRF signaling in the striatum, likely to spark significant interest and future research.

Strengths:

A key strength is the integration of anatomical and functional approaches to demonstrate these projections and assess their impact on target cells, striatal cholinergic interneurons.

Weaknesses:

The nature of the interaction between alcohol and CRF actions on cholinergic neurons remains unclear. Also, further clarification of the ACh sensor used and others is required

<https://doi.org/10.7554/eLife.107145.1.sa2>

Reviewer #3 (Public review):

Summary:

The authors demonstrate that CRF neurons in the extended amygdala form GABAergic synapses onto cholinergic interneurons and that CRF can excite these neurons. The evidence is strong, however, the authors fail to make a compelling connection showing CRF released from these extended amygdala neurons is mediating any of these effects. Further, they show that acute alcohol appears to modulate this action, although the effect size is not particularly robust.

Strengths:

This is an exciting connection from the extended amygdala to the striatum that provides a new direction for how these regions can modulate behavior. The work is rigorous and well done.

Weaknesses:

While the authors show that opto stim of these neurons can increase firing, this is not shown to be CRFR1 dependent. In addition, the effects of acute ethanol are not particularly robust or rigorously evaluated. Further, the opto stim experiments are conducted in an Ai32 mouse, so it is impossible to determine if that is from CEA and BNST, vs. another population of CRF-containing neurons. This is an important caveat.

<https://doi.org/10.7554/eLife.107145.1.sa1>

Reviewer #4 (Public review):

Summary:

This manuscript presents a compelling and methodologically rigorous investigation into how corticotropin-releasing factor (CRF) modulates cholinergic interneurons (CINs) in the dorsal striatum - a brain region central to cognitive flexibility and action selection-and how this circuit is disrupted by alcohol exposure. Through an integrated series of anatomical, optogenetic, electrophysiological, and imaging experiments, the authors uncover a previously uncharacterized CRF⁺ projection from the central amygdala (CeA) and bed nucleus of the stria terminalis (BNST) to dorsal striatal CINs.

Strengths:

Key strengths of the study include the use of state-of-the-art monosynaptic rabies tracing, CRF-Cre transgenic models, CRFR1 reporter lines, and functional validation of synaptic connectivity and neurotransmitter release. The finding that CRF enhances CIN excitability and acetylcholine (ACh) release via CRFR1, and that this effect is attenuated by acute alcohol exposure and withdrawal, provides important mechanistic insight into how stress and alcohol interact to impair striatal function. These results position CRF signaling in CINs as a novel contributor to alcohol use disorder (AUD) pathophysiology, with implications for relapse vulnerability and cognitive inflexibility associated with chronic alcohol intake.

The study is well-structured, with a clear rationale, thorough methodology, and logical progression of results. The discussion effectively contextualizes the findings within broader addiction neuroscience literature and suggests meaningful future directions, including therapeutic targeting of CRFR1 signaling in the dorsal striatum.

Weaknesses:

Minor areas for improvement include occasional redundancy in phrasing, slightly overlong descriptions in the abstract and significance sections, and a need for more concise language in some places. Nevertheless, these do not detract from the manuscript's overall quality or impact.

Overall, this is a highly valuable contribution to the fields of addiction neuroscience and striatal circuit function, offering novel insights into stress-alcohol interactions at the cellular and circuit level, which requires minor editorial revisions.

<https://doi.org/10.7554/eLife.107145.1.sa0>

Author response:

We have outlined a clear plan to revise and strengthen the manuscript by addressing key experimental concerns raised in the public reviews.

Summary of Planned Revisions:

We intend to address the following points through new experiments or additional analyses:

Reviewer #1, Concern 2:

"CRFR1 expression is largely confined to a subpopulation of striatal CINs in rats—Is this also true in mice?"

To address this, we will obtain CRFR1-GFP mice and perform immunohistochemistry for ChAT to assess the overlap between CRFR1-GFP⁺ neurons and CINs in the dorsal striatum.

This will allow us to directly determine whether CRFR1 expression is similarly restricted in mice as it is in rats.

Reviewer #1, Concern 3:

"In rats, ~30% of CINs express CRFR1. Did a similar proportion of CINs in mice respond to CRF application?"

We will revisit and re-analyze our electrophysiological dataset to calculate the percentage of recorded CINs in mice that respond to bath-applied CRF. Our preliminary analysis suggests a higher response rate (>90%), and we will reconcile this with expression data, discuss possible mechanisms (e.g., indirect effects or species-specific differences), and provide a clear explanation in the revised manuscript.

Reviewer #2, Recommendation 5:

"Can the authors quantify the onset delay of optogenetic responses from CRF+ axons onto CINs?"

We initially performed this experiment in a single animal. To strengthen our conclusion of monosynaptic connectivity, we will increase the sample size (additional injections in CRF-Cre mice) and quantify the onset latency of optogenetically evoked responses in CINs.

Reviewer #2, Recommendation 7:

"Are CRFR1+ CINs equally distributed in DMS vs. DLS?"

We will re-analyze existing immunohistochemical images from Figure 4 to compare the density (cells/ μm^2) of CRFR1+ CINs in the dorsomedial vs. dorsolateral striatum. This analysis will help clarify whether there is a regional bias in CRFR1 expression across striatal subdomains.

Reviewer #3, Recommendation 1:

"Test whether CRFR1 mediates the effect of optogenetic stimulation on CIN firing."

We will directly test CRFR1-dependence of optogenetically evoked CIN excitation by applying a CRFR1 antagonist during optical stimulation of CRF+ terminals and evaluating the effect on CIN firing. This will clarify whether the CRF effect is receptor-mediated and strengthen the interpretation of our functional findings.

We may conduct more experiment to address other concerns. These targeted experiments will significantly enhance the rigor and mechanistic insight of our study.

<https://doi.org/10.7554/eLife.107145.1.sa5>